

“I think I can safely say that **nobody** understands quantum mechanics.”

Richard Feynman, 1964

form of the theory, which meant that its equations tended to generate impossible, or infinite, values.

Quantum probabilities

Something of a showman, Feynman had a talent for communicating complex ideas in intuitive ways, and he brought his particular skills to improving the QED theory. In 1948, he formulated a version of the theory that produced meaningful results, with the aid of what became known as Feynman diagrams. These useful graphics illustrated the concepts of electrons and photons moving in both time and space. Crucially, the particles in his diagrams could move not only forward but also backward in time.

Feynman's simple diagrams were in fact a mathematical model in which the probabilities of all possible events could be added up. Many of the probabilities added up to zero and cancelled each other out, leaving answers that made sense in real-world terms—at least, most of the time. The quantum universe is peculiar, and QED is, according to the title of one of Feynman's most popular books, a “strange theory of light and matter.”

In Feynman diagrams, electrons are depicted by straight lines, while photon exchange between electrons is shown by

wavy lines. Looking like doodles at first sight, the diagrams are in reality shorthand for extremely complicated equations. Today, particle physicists use them in a whole range of applications, including calculating possible outcomes of events in particle accelerators such as the Large Hadron Collider in Switzerland. As a result of Feynman's input, QED became one of the most successful theories in physics and was adopted as the model for theories of other basic forces, such as the nuclear force.

Nanotechnology and superfluidity

Feynman made his mark in other key areas of physics, too. He proposed a theory of “partons,” a precursor to the modern concept of quarks (see box); helped to introduce nanotechnology; theorized about the future existence of quantum computing; and explained why, when cooled to absolute zero, helium flows without viscosity (internal friction). In 1986, Feynman formed part of a team investigating the *Challenger* disaster: the NASA space shuttle had exploded seconds after lift-off, killing all seven members of its crew. In a famous public demonstration, Feynman explained how the accident had happened using just a glass of icy water and a piece of rubber.

